TALENTRANK AI

Intelligent Hiring. Explainable Decisions. Production System Design Document

PROJECT: TalentRank AI (Production Recruitment SaaS Platform)
TARGET CLIENT: Staffing Agency in Noida (70 recruiters, 120 employees, 300 reqs/mo, 120 CVs/req)
GITHUB REPO: https://github.com/talentrank-ai/talentrank-ai (Pending Initial Commit)

1. PROJECT OVERVIEW

TalentRank AI is an enterprise-grade AI resume screening and recruitment management platform designed specifically for high-volume staffing agencies operating in Noida and Delhi NCR.

The platform processes contract and permanent IT, banking back-office, and retail recruitment pipelines with approximately 300 open requisitions per month and up to 120 CVs per requisition (36,000 resumes monthly).

Core Mission: Replace arbitrary manual resume keyword matching with objective, explainable, verifiable AI ranking while maintaining human-in-the-loop oversight and strict DPDP Act 2023 compliance.

2. CLIENT PROBLEM & OPERATIONAL BOTTLENECKS

1. High Manual Overhead: 70 recruiters spending over 4.5 hours daily reading unstructured resumes across Java, Python, and banking operations.

2. Subjective & Unreliable Screening: Recruiters frequently reject qualified candidates due to fatigue or keyword mismatches, leading to inconsistent client submissions.

3. Lack of Explainability: Existing legacy ATS solutions generate black-box scores with no evidence, failing client audits.

4. Client Portal Disconnect: Hiring managers receive raw unvetted CVs via email, leaking candidate PII and risking candidate poaching.

3. PROPOSED SOLUTION ARCHITECTURE

TalentRank AI introduces a dual-portal architecture separating Recruiter Operations from Client Hiring Manager Review.

Key Capabilities:

• Gemini AI Scorecard Extraction: Extracts 12 structured criteria from raw JDs into editable evaluation rubrics with customizable weights.

• High-Throughput Bulk CV Ingestion: Parses multi-file PDF and DOCX batches asynchronously with bounded concurrency and duplicate detection.

• Verifiable Quotation Scoring: Every scored dimension is backed by verbatim text extracted from the candidate CV. Missing skills receive a verified score of 0.

• Client Data Isolation & PII Masking: Strict backend authorization ensures client managers only see candidates explicitly submitted to their organization with masked emails and phone numbers.

4. TECHNOLOGY CHOICES & REASONING

• Frontend: React 19, TypeScript, Tailwind CSS, Lucide icons. Delivers a Linear/Stripe-inspired restrained design system free from decorative visual clutter.

• Backend: Node.js, Express, TypeScript, Vite middlewares. Unified server architecture eliminating CORS friction and securing server-side Gemini AI credentials.

• AI Engine: Google Gemini API (gemini-3.8-flash) utilizing structured JSON responseSchema and thinkingLevel tuning for rapid, deterministic inference.

• Document Processing: Mammoth for DOCX and pdf-parse for PDF streams with multimodal OCR fallback.

• Database & Security: In-Memory Firestore-compatible store with full support for Cloud Firestore, enforcing schema constraints and role-based access control.

5. DATABASE SCHEMA & DATA MODEL

Firestore Collections:

1. users: id, email, name, role (recruiter | client), organizationId, avatarUrl, createdAt.

2. organizations: id, name, industry, location, activeRequisitionsCount, contactEmail.

3. requisitions: id, reqNumber, title, organizationId, status, minExp, maxExp, minCtc, maxCtc, maxNoticeDays, rawJd, scorecardId.

4. scorecards: id, requisitionId, jobTitle, location, mustHaveSkills, niceToHaveSkills, weights, version.

5. candidates: id, candidateNumber, name, email (masked for client), phone (masked for client), currentRole, totalExp, skills, extractedCvText, stage, recruiterReviewed.

6. candidateScores: id, candidateId, overallScore, criteria (name, score, evidenceQuotation, confidenceScore, isVerifiedInCvText), strengths, gaps, recommendation.

7. clientSubmissions: id, requisitionId, candidateId, organizationId, status, rejectionReason, feedback.

8. pipelineEvents & auditLogs: immutable event logging tracking user, role, action, target resource, and timestamps.

9. consentRecords & deletionRequests: DPDP Act 2023 compliance records for purpose limitation and right to erasure.

6. AI EXTRACTION ARCHITECTURE

The JD intake pipeline receives raw text or pasted documents and feeds them to Gemini 3.8 Flash with a strict JSON schema. The model extracts 12 structured dimensions:

Job title, location, employment type, min experience, max experience, min CTC, max CTC, max notice period, education, must-have skills, and nice-to-have skills.

Missing requirements are explicitly flagged as unspecified rather than hallucinated. The extracted scorecard is rendered into editable controls allowing recruiters to tweak weights and mandatory thresholds.

7. AI SCORING METHODOLOGY & EVIDENCE VERIFICATION

Normalized Weight Distribution:

• Must-Have Technical Skills: 35% | Experience Relevance: 25% | Nice-To-Have Skills: 15% | Notice Period Fit: 15% | Location & Role Match: 10%

Critical Evidence Rule: For every scored requirement, Gemini must output an exact quotation extracted from the candidate CV text.

Server-Side Verification Engine: The server independently searches the candidate raw text for the returned quotation. If verified, isVerifiedInCvText = true; if not found or fabricated, the criterion is marked unverified with score 0.

8. FAIRNESS SAFEGUARDS & ANTI-BIAS PROTECTIONS

1. Protected Attribute Exclusion: Names, emails, phone numbers, gender, age, marital status, and photographs are stripped prior to scoring.

2. Human-in-the-Loop Gateway: AI recommendations are strictly decision-support tools. Recruiters must open, review, and approve every candidate profile before client submission.

3. Searchability of Low Scores: Low-scoring candidates remain 100% searchable in tables and filters to ensure diverse consideration.

4. Delivery-Head Fairness Review: Weekly randomized sample audits human override rates (e.g. 8.3%) and divergence factors.

9. PRIVACY, SECURITY & DPDP ACT 2023 COMPLIANCE

• Purpose Limitation: Candidate CVs are processed strictly for recruitment evaluation under explicit consent records.

• Client PII Masking: Client hiring managers only receive masked emails (*****@masked.talentrank.ai) and masked phone numbers (+91 ***** *****).

• Right to Erasure Workflow: Executes data subject deletion requests by permanently purging raw CV text, phone, and email while preserving minimal audit IDs.

• Tamper-Evident Audit Logging: Immutable logging of profile views, stage transitions, and client actions.

10. TESTING STRATEGY & QUALITY ASSURANCE

Automated test suite (src/test/runAllTests.ts) executes and validates all 14 mandatory functional scenarios:

1. JD scorecard extraction schema | 2. PDF parsing | 3. DOCX parsing | 4. Duplicate detection | 5. Scoring calculation | 6. Verifiable quotations | 7. Missing evidence handling | 8. Protected attribute exclusion | 9. Recruiter-only access | 10. Client data isolation | 11. Recruiter review gate | 12. Mandatory client rejection reason | 13. Pipeline transitions | 14. DPDP consent & erasure.

11. SCOPE LIMITATIONS & FUTURE IMPROVEMENTS

• Intentionally Excluded: Direct WhatsApp/Email dispatch (simulated in-app to avoid spam traps); External third-party calendar synchronizations.

• Future Roadmap: Multilingual resume parsing for regional Indian languages; Automated interview video transcript analysis; Advanced compensation benchmarking.